PRODUCT USER MANUAL

Atlantic - European North West Shelf - Ocean Physics Analysis and Forecast Product (Low Resolution)

NWSHELF_ANALYSISFORECAST_PHY_LR_004_001

Issue: 1.0

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2025/05	All	Initial version after Met Office reintroduction of this product	J. Tinker, T. Brüning	Copernicus Marine Product Management

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GLOSSARY AND ABBREVIATIONS

MFC	Monitoring and Forecasting Centre
NWS	North-West Shelf
TAC	Thematic Assembly Centre
IOC	Intergovernmental Oceanographic Commission of UNESCO
NRT	Near Real Time
PUM	Product User Manual
QUID	Quality Information Document
CMEMS	Copernicus Marine Environment Monitoring Service
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
MDS	Marine Data Store
FORECAST	An assessment of the future state of the ocean variables made using numerical models.
HINDCAST	An assessment of the past state of the ocean variables made using numerical models with or without data assimilation. Simulations and analyses are hindcasts.
SIMULATION	A hindcast done without data assimilation.
ANALYSIS	A hindcast adjusted with data assimilation
SSS	Sea Surface Salinity
SSC	Sea Surface Current
SSH	Sea Surface Height
SST	Sea Surface Temperature

Zonal Velocity	West to East component of the horizontal velocity vector
Meridional Velocity	South to North component of the horizontal velocity vector

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us.

1) INTRODUCTION

a) Summary

This document is the user manual for the North West-European Shelf analysis and forecast product:

NWSHELF_ANALYSISFORECAST_PHY_LR_004_001

This products is generated by a coupled physics-biogeochemistry forecasting system. The corresponding biogeochemistry product is **NWSHELF_ANALYSISFORECAST_BGC_004_002**, which is described here.

Every day:

- the time series (analysis for 2 days before present and 7 days forecast) is updated at the end of the operational production process
- old forecasts are deleted from the MDS every day

The product is published on and available through the MDS after automatic and human quality controls. Downloaded files are in NetCDF format and follow CF-1.7 convention.

The analysis and forecasting system is described in the Quality Information Document (QUID):

https://documentation.marine.copernicus.eu/QUID/CMEMS-NWS-QUID-004-001.pdf

More detailed information can be obtained from the Copernicus Marine Service Team.

b) History of changes

Date	Description of changes and impacted product
November 2025	Re-introduction of the NWS physical analysis and forecasting product by the UK Met Office including a 7-day forecast

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

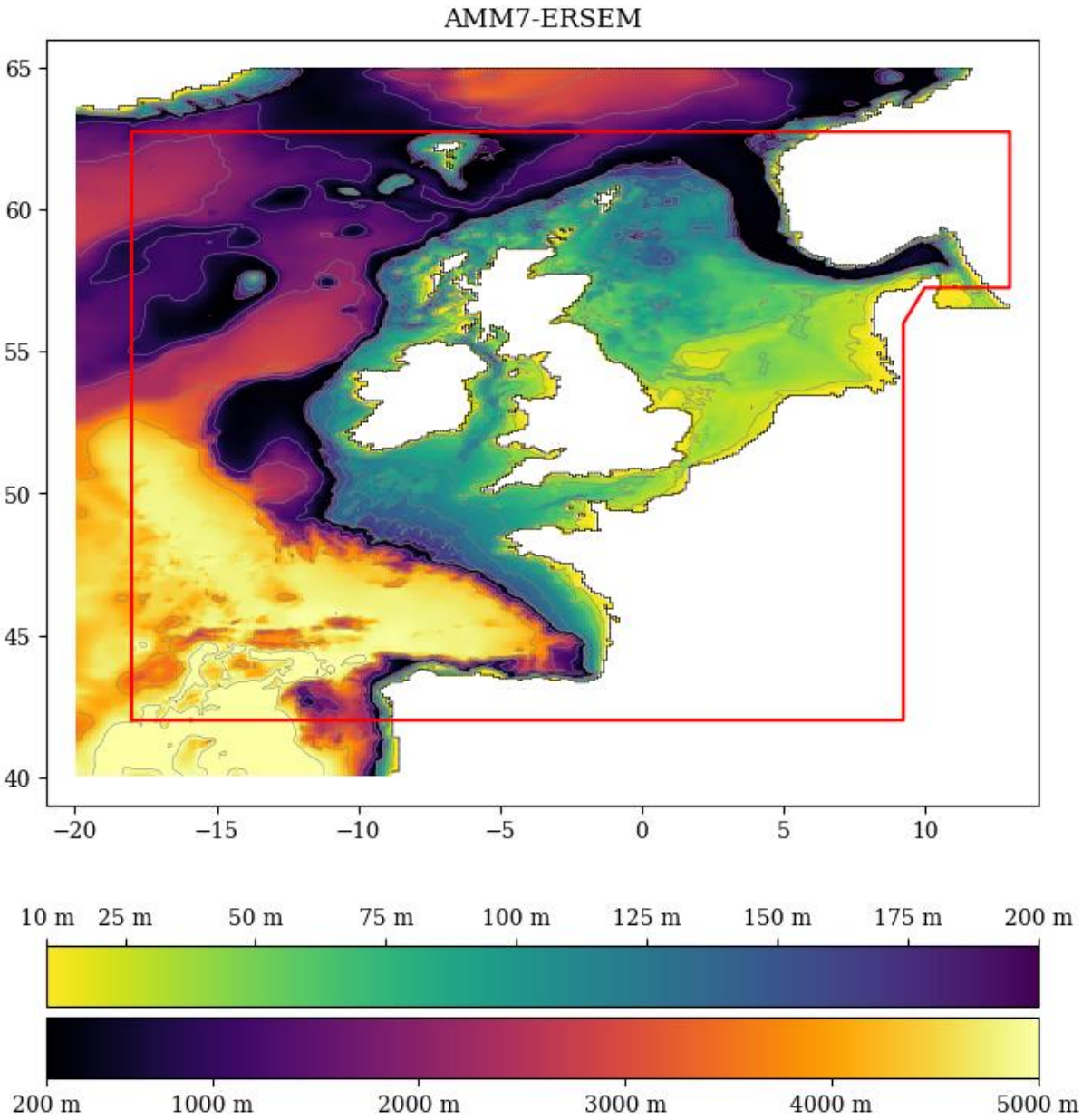
Product Lines	NWSHELF_ANALYSISFORECAST_PHY_LR_004_001	
Geographical coverage	20°W to 13°E; 40°N to 65°N. The outermost model grid points at the Atlantic and Baltic boundaries are masked. Baltic boundary: all the grid points in the Baltic and the Kattegat South of 57.25°N are masked. Atlantic boundaries: grid points north of 64°18'48" N; west of -18°19'12" E and south of 40°43'58" N are masked (outermost 10-grid points).	
Variables	• Potential temperature • Salinity • Sea surface height • Horizontal velocity (eastward and northward components) • Bottom Temperature • Mixed Layer depth	
Product Type	Near Real Time - Analysis	Near Real Time - Forecast
Available time series	Two years before present	7 days forecast
Temporal resolution	Daily means (25-hour mean[1]);	Daily means (25-hour mean[1]);
Update Frequency	Daily	Daily
Target delivery time	daily at 12 pm (noon) UTC	
Horizontal resolution	1/9° longitude x 1/15° latitude	
Number of vertical levels	24	
Format	NetCDF4 - CF1.7	
Delivery mechanism	Copernicus Marine Toolbox	

[1] Our daily mean products are the mean of 25 instantaneous means, including the previous and following midnight values – this effectively removes the effect of the tide in the data.

b) Production System Description

The operational North-West-European Shelf analysis and forecast products are generated by a coupled physical-biogeochemical forecasting system at 1/15° (longitude) and 1/9° (latitude It provides a 7-day forecast and two analysis days of 2D/3D (PHY_LR_004_001) and (BGC_004_002) data updated daily. This product includes daily mean detided files. A rolling archive of 2 years of analysis is available.

NORTH WEST EUROPEAN SHELF (20°W - 13°E; 40°N - 65°N) :
1/15° longitude x 1/9° latitude, regular grid, 297x 375

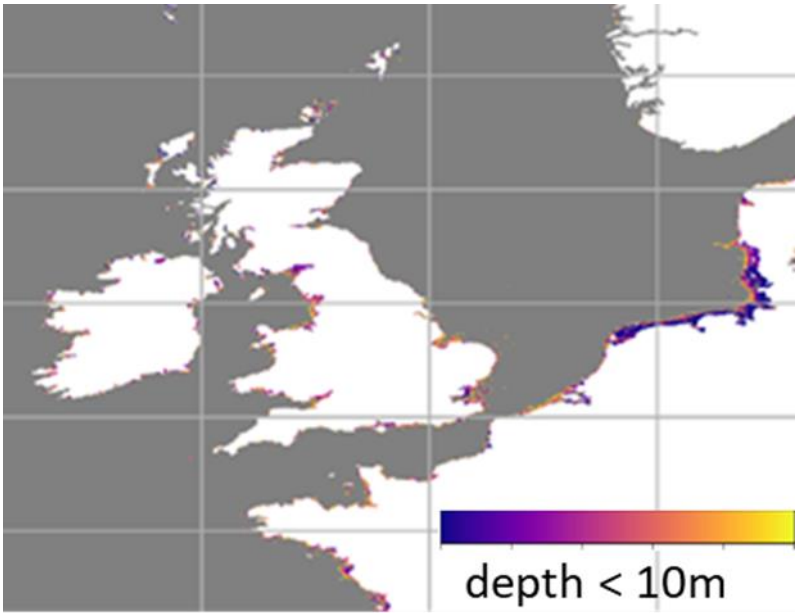


NWS bathymetry (m) of the AMM7 model domain, with the red line showing the product domain.

The domain extends beyond the continental shelf in order to place the model's boundary region in the deep waters of the adjacent North-East Atlantic, but the focus region for the model comprises open waters of the shelf seas, i.e. (using UK terminology) the North Sea, Irish Sea, English Channel, Celtic Sea and Bay of Biscay.

The model is a nested regional ocean model configuration, coupled with the biogeochemical model. PHY and BIO products are delivered on an identical horizontal grid.

A **minimum depth of 10m** is imposed in the ocean model for considering the sea level tidal excursion. Products below the surface in areas where the bathymetry is less than 10m could be affected by this modelling choice.



1) What is done:

We take every bathymetric grid point < 10 m in the domain and reassign it with a depth of 10 m, coloured point in the figure above. These grid points are interpolated back to their original depth before delivering the file to the CMEMS catalogue. Areas such as the Wadden Sea are particularly affected (see below reference for a case study in the Wadden sea).

2) Why it is done:

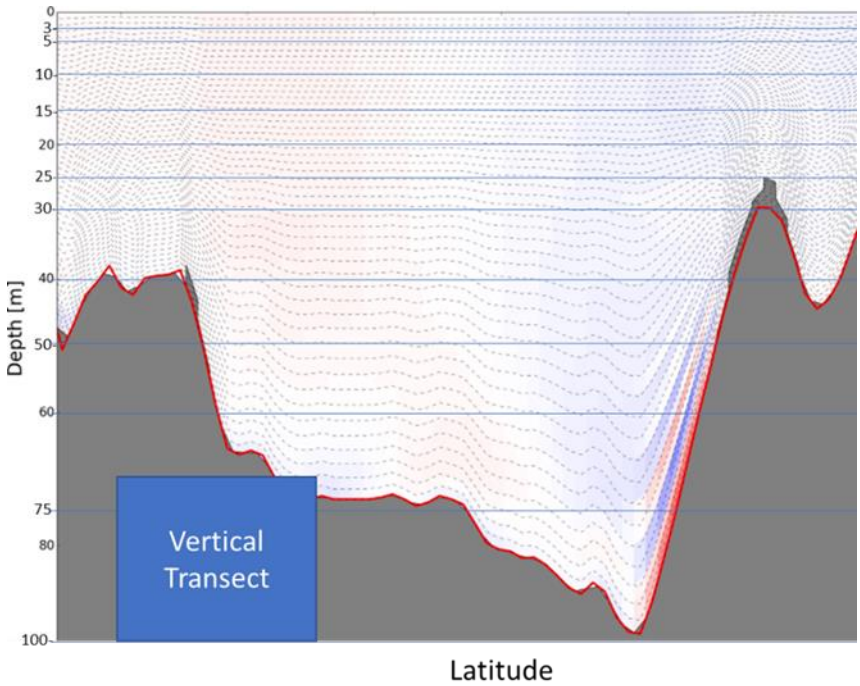
This is because version 3.6 of NEMO does not have wetting and drying capability. If the bathymetry were not modified, it would mean that with the large tides that exist in this domain the model would not be able to handle the case when the tide goes out.
The minimum depth prevents such a situation from occurring and keeps the model as a whole stable.

3) What is the likely effect.

The prorogation of tides and surges are inherently related to the water depth. The speed is proportional to $\sqrt{gH}$ where H is the water depth and thus as the model water depth is deeper than reality the prorogation will be incorrect.
In cases where the seabed is exposed then this process is completely absent from the model by design (there will be water in the model where in reality there is none). It is thus not recommended to the user to directly use the model fields in regions where there are extensive areas of bathymetry less than 10 m in depth. If a model is to be nested within this product, then it is recommended that the boundaries of the nested domain are sufficiently far from such areas to reduce their impact.

4) Wetting and Drying is implemented in NEMO 4.

This specific problem is addressed with the wetting and drying module in NEMO 4 which will be part of future development of this operational Copernicus product.
The interested reader is directed to the recent paper on its development:
https://doi.org/10.1016/j.ocemod.2020.101708
Enda O'Dea, Michael J. Bell, Andrew Coward, Jason Holt, Implementation and assessment of a flux limiter based wetting and drying scheme in NEMO, Ocean Modelling, Volume 155,2020.

Horizontal grid	~7km (1/9°longitude x 1/15°latitude).
Vertical grid	The products are interpolated for users' convenience, into 24 standard IOC geopotential levels: [surface, 3, 10, 15, 20, 30, 50, 75, 100, 125, 150, 200, 250,300, 400, 500, 600, 750, 1000, 1500, 2000, 3000, 4000, 5000 m]. The surface level is not interpolated, it is the first model level. It has a thickness of 1m where the bathymetry is deeper than 50m. Its thickness is less than 1m where the bathymetry is shallower than 50m.  *Vertical transect: it shows the differences between the model native vertical grid (dotted line) and the CMEMS products interpolated at standard geopotential depths (blue lines). The model bathymetry (red lines) doesn't match exactly the bathymetry (shaded grey area) due to numerical constrains.* The model native vertical grid has Hybrid S-σ-z-coordinates. The terrain-following coordinates system uses a stretching function to maintain near uniform vertical resolution at the surface. The number of S-σ-z levels is 51 almost everywhere in model domain. It means that

	the thickness of the model vertical grid cells is smaller in shallow area compared to the deep part of the basin. The thickness of the model native vertical grid varies from 0.3m (in areas with a bathymetry < 50m) to 99m (where the bathymetry is > 4000m). The bottom temperature (bottomT) is output directly from the native model, it is not interpolated. All other CMEMS NWS-MFC products are vertically interpolated from the 51 native terrain following coordinates to fixed depths.
System Name	AMM7
Model version	NEMO 3.6 (coupled to biogeochemical model ERSEM22.11 via FABM)
Coupling	One-way on-line coupling: physics forcing the biogeochemistry. Coupling frequency: at every model time step.
Bathymetry	GEBCO 1 arc-minute data together with a variety of other local data sources, supplied by the North West-European Shelf Operational Oceanographic System (NOOS) partners.
Initial conditions	The model run started on 01/01/2017 from the previous version of NEMO-ERSEM forecasting system.
Atmospheric forcings	Met Office Global Unified Model (MetUM)
Tidal constituents	Tidal forcing is included both on the open boundary conditions via a Flather radiation boundary conditions (Flather, 1976) and through the inclusion of equilibrium tide from a tidal model of the North-East Atlantic (Flather, 1981). 15 tidal constituents: M_2, S_2, N_2, K_2, K_1, O_1, P_1, Q_1, M_4, MS_4, L_2, T_2, S_1, $2N_2$, MU_2, NU_2.
Rivers	Daily timeseries of river discharge (as well as nutrient loads (nitrate, phosphate, silicate, ammonia), alkalinity (total alkalinity, bioalkalinity, dissolved organic carbon), and oxygen) from 1991-2017. Timeseries data are produced from an updated version of the river dataset used in Lenhart et al. (2010) combined with climatology of Daily discharge data from the Global River Discharge Data Base (Vörösmarty et al., 2000) and from data prepared by the Centre for Ecology and Hydrology as used by Young and Holt (2007). Interannual changes in the series are small and so the 2017 time series is repeated for subsequent years.
Lateral boundary condition	• Atlantic (T, S, SSH, barotropic u and v): Met Office Operational ORCA12 forecast implementation. • Baltic (T, S, barotropic u and v): • BALTICSEA_ANALYSISFORECAST_PHY_003_006.
Data Assimilation scheme	3D Var FGAT. NEMOVAR.
Assimilated observations	• In situ and satellite (L2/L3 products) SST. • satellite SLA from CMEMS SL-TAC. • temperature and salinity subsurface profiles from GTS and CMEMS INS-TAC.

c) Dataset details

NWSHELF_ANALYSISFORECAST_PHY_LR_004_001	
Daily (25-hour[1]) mean **(P1D-m)** fields of: • Potential temperature, salinity, zonal and meridional velocity (3D) • Sea Surface Height (SSH), sea bottom temperature and 2D Mixed layer depth • Sea Surface Temperature (SST) • Sea Surface Salinity (SSS) • Sea Surface Current (SSC)	
Datasets	Variables name in the NetCDF file [Unit] Long_name Standard_name
cmems_mod_nws_phy-**uv**_anfc_7km-3D_P1D-m	**vo** [ms^{-1}] Northward Current Velocity in the Water Column northward_sea_water_velocity
	uo [ms^{-1}] Eastward Current Velocity in the Water Column eastward_sea_water_velocity
cmems_mod_nws_phy-**tem**_anfc_7km-3D_P1D-m	**thetao** [degrees_C] Sea Water Potential Temperature sea_water_potential_temperature
cmems_mod_nws_phy-**sal**_anfc_7km-3D_P1D-m	**so** [1e-3] Sea Water Salinity sea_water_salinity
cmems_mod_nws_phy-**ssh**_anfc_7km-2D_P1D-m	**zos** [m] Sea surface height above geoid sea_surface_height_above_geoid
cmems_mod_nws_phy-**bottomT**_anfc_7km-2D_P1D-m	**bottomT** [degrees_C] Sea floor potential temperature sea_water_potential_temperature_at_sea_floor
cmems_mod_nws_phy-**mld**_anfc_7km-2D_P1D-m	**mlotst** [m] ocean mixed layer thickness defined by density ocean_mixed_layer_thickness_defined_by_sigma_theta
cmems_mod_nws_phy_anfc_7km-3D_**static**	**mask** Land-sea mask sea_binary_mask
	depth [m] depth depth
	deptho [m] Bathymetry sea_floor_depth_below_geoid
	mdt[m] Mean Dynamic Topography sea_surface_height_above_geoid
	deptho_lev_interp deepest wet grid point of the interpolated 3D fields model_level_number_at_sea_floor

The model **bathymetry, Land-Sea mask** and Mean Dynamic Topography **(MDT)** of this product is also used by the product NWSHELF_MULTIYEAR_BGC_004_002, as both products are produced by a coupled physical-biogeochemical system.

d) Additional information on some parameters

mlotst [m]	ocean_mixed_layer_thickness_defined_by_sigma_theta. It is the depth where the density increase compared to density at 10 m depth corresponds to a temperature decrease of 0.2°C in local surface conditions (θ10m, S10m, P0= 0 db, surface pressure)
zos [m]	sea_surface_height_above_geoid. The geoid is a surface of constant geopotential with which mean sea level would coincide if the ocean were at rest. The parameter "zos" is the difference between the actual sea surface height at any given time and place, and that which it would have if the ocean were at rest.
bottomT (degrees_C)	sea_water_potential_temperature_at_sea_floor is output directly from the native model; it is not interpolated. It means that bottom and the lowest wet point of the 3D temperature dataset could be different.
deptho_lev_interp	Deepest wet grid point of the Interpolated 3D fields.

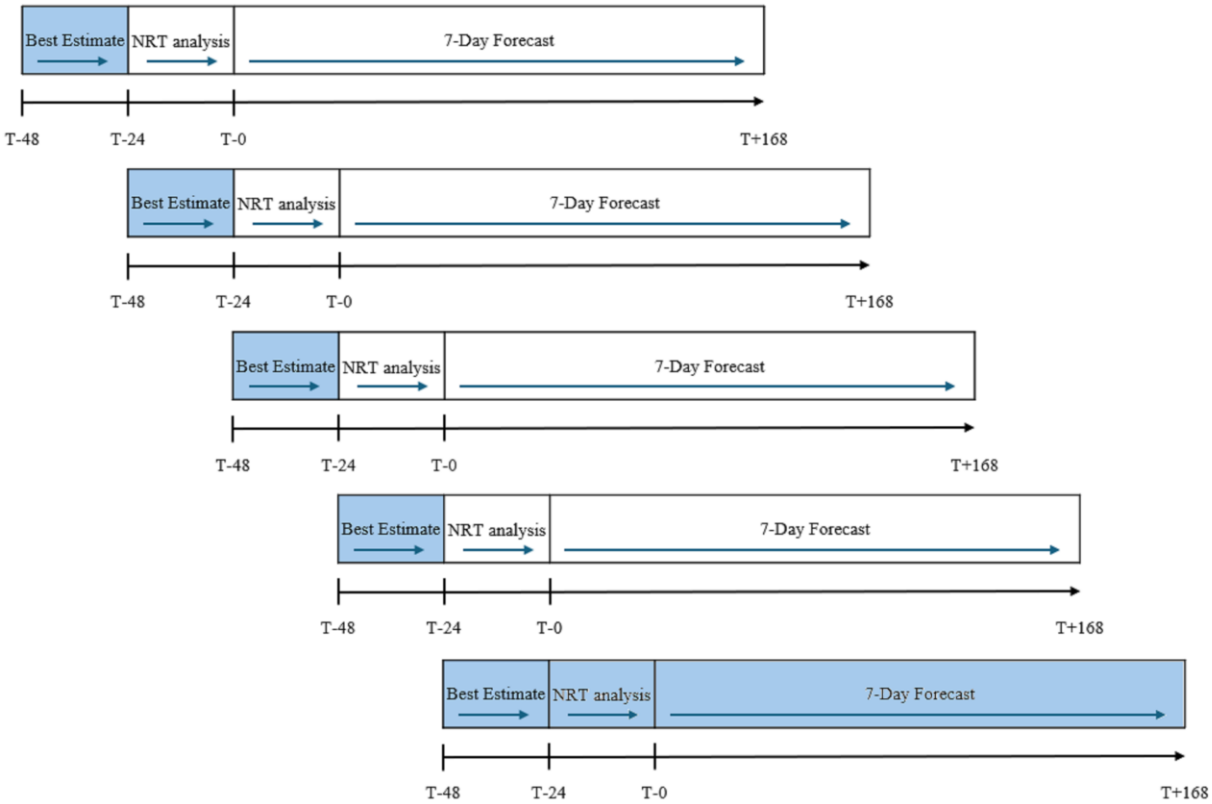
e) Processing information

Product disseminating schedule:

Type of product (forecast/ hindcast)	Forecast run (base time)	Product temporal coverage (hours)	Product disseminating scheduled time	Dissemination day
Forecast	00 UTC	0 to 168	until 12 UTC	Every day
NRT - Analysis	00 UTC	-24 to 0	until 12 UTC	Every day
Best estimate - Analysis	00 UTC	-48 to -24	until 12 UTC	Every day

f) Catalogue updating

The time coverage of the catalogue data spans from two years before present to present plus 7 days of forecast.



Each day the [T-48h, T-24h] and [T-24h, T+00h] analyses are provided along with 7 days forecast for this product (with the first day of the forecast being for the day of production).
The delivery usually happens between 6:40 AM and 10:00 AM, a couple of hours before the target delivery time. The previous day 'NRT analysis' and '7-day of forecast' are deleted at the beginning of this process. The catalogue data from 2-year before present to present minus 24 hours is taken from T-48h to T-24h of each run (Best Estimate – Analysis).

g) Time averaging

Dataset frequency	Time averaging	Forecast lead time
daily mean values	Values are calculated as means of 25-hour instantaneous values, starting at midnight (UTC) and finishing on the following midnight to remove both diurnal and tidal cycles.	7-day (168-hr)

h) Product use and impact

Specialised marine physics analysis and forecast product for the Northwest Shelf region. Uses include marine navigation, emergency response (search and rescue/pollution), coastal flood mitigation, health and ecosystems monitoring, energy and net zero. Applications are developed for government policy, industrial decision making and academic research. The NWS products provide high resolution data and forecasts for ocean state in the tidally dominated waters of the Northwest Shelf and are designed to provide state of the art information specific to this region.

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

4.1) FILE SIZE

NWSHELF_ANALYSISFORECAST_PHY_LR_004_001		
DATASET NAME	**NAME OF FILE**	**DIMENSION [MB]**
cmems_mod_nws_phy-uv_anfc_7km-3D_P1D-m	metoffice_foam1_amm7_NWS_CUR_b{{yyyymmdd}}_dm{{yyyymmdd}}.nc	2.2
cmems_mod_nws_phy-tem_anfc_7km-3D_P1D-m	metoffice_foam1_amm7_NWS_TEM_b{{yyyymmdd}}_dm{{yyyymmdd}}.nc	1.3
cmems_mod_nws_phy-sal_anfc_7km-3D_P1D-m	metoffice_foam1_amm7_NWS_SAL_b{{yyyymmdd}}_dm{{yyyymmdd}}.nc	1.0
cmems_mod_nws_phy-bottomt_anfc_7km-2D_P1D-m	metoffice_foam1_amm7_NWS_BED_b{{yyyymmdd}}_dm{{yyyymmdd}}.nc	0.1
cmems_mod_nws_phy-mld_anfc_7km-2D_P1D-m	metoffice_foam1_amm7_NWS_MLD_b{{yyyymmdd}}_dm{{yyyymmdd}}.nc	0.1
cmems_mod_nws_phy-ssh_anfc_7km-2D_P1D-m	metoffice_foam1_amm7_NWS_SSH_b{{yyyymmdd}}_dm{{yyyymmdd}}.nc	0.1
cmems_mod_nws_phy_anfc_7km-3D_static	NWS-MFC_004_001_static.nc	12

4.2) STRUCTURE OF FILES

An Example of the header of output NetCDF files is inserted in **annex**.

Remember: scale_factor & add_offset / missing_value / land mask :

Real_Value = (Display_Value X scale_factor) + add_offset
The missing value for this product is: -32768s
Land and sea-ice masks are equal to "_FillValue" (see variable attribute on NetCDF file).

5) REFERENCES

Quality Information Document
https://documentation.marine.copernicus.eu/QUID/CMEMS-NWS-QUID-004-001.pdf

Flather R.A., (1976). A tidal model of the north west European continental shelf. Memoires de la Societe Royale de Sciences de Liege,6,141-164.

Flather R.A., (1981). Practical surge prediction using numerical models. In: Peregrine DH (ed) Floods due to high winds and tides. Academic Press, London New York.

Lenhart H.-J., Mills D.K., Baretta-Bekker H., van Leeuwen S.M., van der Molen J., Baretta J.W., Blaas M., Desmit X., Kühn W., Los H.J., Ménesguen A., Neves R., Proctor R., Ruardij P., Skogen M.D., Vanhoutte-Brunier A., Villars M.T., Wakelin S., (2010). Predicting the consequences of nutrient reduction on the eutrophication status of the North Sea, Journal of Marine Systems, Vol 81,148-170.

Vörösmarty J., Green P., Salisbury J., Lammers R. B., (2000). Global Water Resources: Vulnerability from Climate Change and Population Growth, Science, 289, 284–288, https://doi.org/10.1126/science.289.5477.284

Young E.F, Holt J.T., (2007). Prediction and analysis of long-term variability of temperature and salinity in the Irish Sea. Journal of Geophysical Research, 112 (C1). Art. No. C01008. https://doi.org/10.1029/2005JC003386

6) ANNEX

```
netcdf metoffice_foam1_amm7_NWS_TEM_b20250526_dm20250524 {
dimensions:
        depth = 24 ;
        time = UNLIMITED ; // (1 currently)
        latitude = 375 ;
        longitude = 297 ;
variables:
        double time(time) ;
                time:axis = "T" ;
                time:standard_name = "time" ;
                time:data_time = 86400.f ;
                time:long_name = "Validity time" ;
                time:calendar = "Gregorian" ;
                time:units = "seconds since 1970-01-01T00:00:00Z" ;
        float latitude(latitude) ;
                latitude:axis = "Y" ;
                latitude:standard_name = "latitude" ;
                latitude:units = "degrees_north" ;
                latitude:nav_model = "grid_T" ;
                latitude:long_name = "latitude" ;
                latitude:valid_min = -90.f ;
                latitude:valid_max = 90.f ;
        float longitude(longitude) ;
                longitude:axis = "X" ;
                longitude:standard_name = "longitude" ;
                longitude:units = "degrees_east" ;
                longitude:nav_model = "grid_T" ;
                longitude:long_name = "longitude" ;
                longitude:valid_min = -180.f ;
                longitude:valid_max = 180.f ;
        short thetao(time, depth, latitude, longitude) ;
                thetao:_FillValue = -32768s ;
                thetao:missing_value = -32768s ;
                thetao:scale_factor = 0.001f ;
                thetao:add_offset = 25.f ;
                thetao:standard_name = "sea_water_potential_temperature" ;
                thetao:long_name = "Sea Water Potential Temperature" ;
                thetao:units = "degrees_C" ;
                thetao:valid_min = -30000s ;
                thetao:valid_max = 30000s ;
        float depth(depth) ;
                depth:axis = "Z" ;
                depth:standard_name = "depth" ;
                depth:units = "m" ;
                depth:positive = "down" ;
                depth:long_name = "depth" ;

// global attributes:
                :Conventions = "CF-1.7" ;
                :title = "daily-mean potential temperature (3D)" ;
                :references = "http://marine.copernicus.eu/" ;
```

:institution = "UK Met Office" ;
:contact = "servicedesk.cmems@mercator-ocean.eu" ;
:netcdf-version-id = "netCDF-4" ;
:creation_date = "2025-05-26T07:06:36Z" ;
:source = "PS-OS 47, AMM-FOAM 7 km (tidal) NEMO v3.6_FABM-ERSEM v19.04_NEMOVAR v6" ;
:credit = "E.U. Copernicus Marine Service Information (CMEMS)" ;
:licence = "http://marine.copernicus.eu/services-portfolio/service-commitments-and-licence/" ;
:product = "NWSHELF_ANALYSISFORECAST_PHY_LR_004_001" ;
:history = "" ;
:forcing_data_source = "UKMO Unified Model (MetUM) Atmospheric model Coupled-NWP (applied as direct fluxes); UKMO global ORCA12 ocean model; UKMO GLOWave; CMEMS-BALTIC; climatological rivers" ;
}